Exploring Constructive Approaches to the Umans–Wang (α, β)-Divisor Conjecture:

A Survey of Construction Strategies, Empirical Obstructions, and Lattice-Based Optimality Results

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Provenance and Status

This document was produced through an extended, iterative conversation between a human collaborator and Claude, an AI model developed by Anthropic. Real errors were introduced and subsequently caught during the investigation: a sign error in an early continued-fraction computation; a degenerate zero-valued covering term that initially passed verification spuriously; a random-seed bug making nominally independent trials identical; and a floating-point precision failure that produced confident but wrong numerical results before diagnosis. These are disclosed because they are relevant to how the results should be weighed.

This work has **not** been peer-reviewed, has not been checked by a domain expert in number theory or computational complexity, and has not been submitted to any venue. The three theorems are carefully checked but not independently verified. The non-degeneracy lemma (Lemma 8.3) rests on one cited classical fact (Lemma 8.2) rather than a fully self-contained derivation. The numerical results in Section 9 characterise the empirical performance of our specific lattice-reduction algorithm relative to a target that Theorem 9.1 already establishes unconditionally; they should not be read as evidence that practical performance converges to 3/2 independently of that theorem. The literature search supporting novelty claims was targeted, not exhaustive.

Abstract

Umans and Wang [1] introduced the (α, β)-Divisor Conjecture: whether there exist sets $S, T \subseteq \mathbb{N}$, each of cardinality at most n^β, with elements of magnitude at most $\exp(n^\alpha)$, such that every integer $i \leq n$ divides some difference $s - t$ with $s \in S$, $t \in T$. A positive resolution with $\alpha, \beta < 1/2$ would yield the first improvement in over a decade to the best known algorithms for polynomial factorisation over finite fields and for deterministic integer factorisation.

We document a systematic exploration of construction strategies spanning elementary number theory, the probabilistic method, continued fractions and Diophantine approximation, discrete logarithms, the geometry of numbers, and classical addition-chain theory. We report three new, fully proved results:

(i) an exact characterisation of the optimal trade-off achievable by a natural family of *hypercube lattice* constructions (Theorem 8.1);

(ii) a strictly stronger *full-line achievability* theorem showing a related star/chain construction reaches the trivial bound $\alpha + \beta = 1$ continuously (Theorem 9.1);

(iii) a unifying theorem proving the star/chain structure is the global optimum of the entire two-parameter family of blocked hypercube–star hybrids, via the one-line inequality $2^d - 1 \geq d$ (Theorem 9.3).

We construct explicit, verified witnesses approaching the exponent of the best previously known algorithm [2], and examine carefully what those experiments do and do not establish. The (α, β)-Divisor Conjecture remains open.

Contents

1 The Umans–Wang Conjecture

In November 2025, Umans and Wang [1] posed a number-theoretic conjecture with a striking algorithmic consequence. The current best algorithm for factoring polynomials over finite fields runs in time roughly $n^{3/2}$, a barrier standing since Kedlaya–Umans [2]; the conjecture, if true with the right parameters, would push this exponent to $n^{4/3}$.

Definition 1.1 ((α, β)-covering pair)**.** *For $n \in \mathbb{N}$ and $\alpha, \beta > 0$, a pair (S, T) of sets of positive integers is an* (α, β)-covering pair for n *if $|S|, |T| \le n^{\beta}$, every element of $S \cup T$ is at most $\exp(n^{\alpha})$, and for every $i \in \{1, \dots, n\}$ there exist $s \in S$, $t \in T$ with $i \mid s - t$.*

Definition 1.2 (n-divisor property [1, Def. 3.1])**.** *A set $A \subseteq \mathbb{N}$ satisfies the* n-divisor property *if for every $i \in \{1, \dots, n\}$ there exists $a \in A$ with $i \mid a$.*

The *Strong (α, β)-Divisor Conjecture* asks whether covering pairs exist with $\alpha, \beta < 1/2$. Two trivial constructions span the easy ends: $S = \{1, \dots, n\}$, $T = \{0\}$ gives $\alpha \approx 0$, $\beta = 1$; $S = \{\mathrm{lcm}(1, \dots, n)\}$, $T = \{0\}$ gives $\beta = 0$, $\alpha \approx 1$. Both sit on the line $\alpha + \beta = 1$.

2 Elementary Constructions and the Trivial Floor

Bucketing constructions — partitioning $[1, n]$ into m groups and forming S from group LCMs — achieve $\beta = \log(m)/\log(n)$ and $\alpha = 1 - \beta + o(1)$, tracing the trivial line exactly. Arithmetic-progression variants, motivated by Proposition 3.4 of [1] (which reduces the conjecture to a single-AP question), gave the same trade-off in every configuration tested. Greedy combinatorial search over smooth-number candidates similarly landed at or above $\alpha + \beta \approx 1$ across all problem sizes, consistent with the combinatorial impossibility result of Lemma 6.3 in [1].

3 An Exact Construction via the Chinese Remainder Theorem

Lemma 6.2 of [1] gives an exact algebraic characterisation: an arithmetic progression covers every prime $p \le n$ if and only if there is a factorisation of the primorial $U = \prod_{p \le n} p$ together with integers b, c satisfying

$$c \cdot S + b \equiv 0 \pmod{U},$$

where S is a sum of Chinese Remainder idempotents determined by the factorisation. For fixed factorisation, finding the smallest (b, c) pair reduces to continued-fraction approximation of S/U. After correcting an initial sign error ($b \equiv +cS$ instead of $-cS$) and a degenerate zero-term bug (a covering term equal to 0, trivially divisible by every prime but mathematically meaningless), the implementation produced verified covering progressions with magnitudes well below the generic $\sqrt{U}$ baseline.

A scaling study from $n = 10$ to $n = 150$ showed real but logarithmic magnitude savings consistent with the metric theory of continued fractions (Khinchin [6], Postnikov [7]): the typical partial quotient in the continued fraction of a "generic" number grows like $O(\log k)$, so the savings are real but do not change the leading-order exponent.

4 The Discrete-Logarithm Sieve

For a fixed base g and prime p, one has $p \mid g^i - 1$ if and only if $\mathrm{ord}_p(g) \mid i$. Taking $S = \{g, g^2, \ldots, g^L\}$ and $T = \{1\}$ therefore covers any prime whose multiplicative order under g divides some $i \leq L$. Direct computation of orders across primes up to several hundred showed coverage growing essentially linearly with L, with no anomalous smoothness for any fixed small base — consistent with known results on the average multiplicative order of a fixed integer mod p. A hybrid construction pre-filtering smooth-order primes before handing the residual to the continued-fraction search was implemented and found to have a strictly worse combined (α, β) trade-off than the continued-fraction method alone in every configuration tested.

5 The Probabilistic Method

5.1 First moment

For S, T chosen as uniform random subsets of size K in a large range, the expected number of uncovered divisors $i \leq n$ is $\sum_{i=1}^{n} \exp(-K^2/i)$. A union bound shows this expectation falls below 1 once $K \gtrsim \sqrt{n \log n}$, giving a threshold $\beta \to 1/2$ as $n \to \infty$. Numerical verification across $n \in \{50, 100, 500, 1000\}$ confirmed convergence of the implied β to $1/2$ exactly, independently recovering the trivial construction's floor via a purely probabilistic argument.

5.2 Structured randomness

Sampling S and T as random arithmetic progressions rather than uniform sets was tested directly and found to perform no better, and measurably worse, than uniform sampling. The first-moment calculation depends only on marginal probabilities that a single pair (s, t) satisfies divisibility, which structured sampling does not improve; AP-structured differences are less varied than uniform ones, which can only hurt coverage.

5.3 Dispersion method

An attempt to apply Linnik's dispersion method [5] (the second-moment escalation used to solve binary additive problems such as the Titchmarsh divisor problem) found that the events "i is uncovered" and "j is uncovered" are *positively* correlated: failing to cover one divisor increases the probability of failing to cover another. This is the wrong sign for Janson-inequality refinements to help. We computed this for one representative pair $(i, j) = (40, 60)$ analytically and by simulation; we did not compute the aggregate dependency term $\Delta = \sum_{i,j} \mathrm{Cov}(X_i, X_j)$ required for a fully rigorous conclusion, so this should be read as a plausible mechanistic explanation rather than a proof.

6 The Alteration Method

Deliberately undershooting the random-construction threshold by roughly 30% and patching each leftover uncovered divisor with one targeted extra element achieved a real 20–25% reduction in total construction size relative to forcing full coverage outright. However, composing this with the exact CRT/continued-fraction construction of Section 3 revealed a structural mismatch: the continued-fraction construction succeeds with certainty for any valid factorisation (by Bézout's identity), leaving no probabilistic near-miss for the alteration method to patch. Across hundreds of trials at n up to 800, the construction achieved complete coverage in every instance.

7 Generalised Progressions and the Lattice Method

Conjecture 3.3 of [1] requires S and T to be sums of several arithmetic progressions. Testing the two-progression case via Lemma 6.1's combination formula found it strictly worse than the single-progression baseline: the count overhead of the combination formula was not offset by a sufficient reduction in magnitude.

Reformulating as a joint optimisation revealed the problem reduces to a lattice question. With k independent common differences $b_1, \ldots, b_k$ and shared offset B, the covering condition collapses via CRT idempotents to

$$B + \sum_{i=1}^{k} b_i W_i \equiv 0 \pmod{U},$$

where $W_i \in \mathbb{Z}/U\mathbb{Z}$ are constants determined by the prime partition. This is a rank-$(k+1)$ lattice of covolume U, and Minkowski's First Theorem guarantees a nonzero point with all coordinates bounded by $C \cdot U^{1/(k+1)}$.

8 A Proved Result: Optimality of Hypercube Lattice Constructions

8.1 Theorem 1

Theorem 8.1 (Optimality of hypercube constructions)**.** *For the hypercube construction with k binary dimensions indexed by $\varepsilon \in \{0,1\}^k$ (i.e. $S = \{B + \sum_i \varepsilon_i b_i : \varepsilon \in \{0,1\}^k\}$),* *the rank-$(k+1)$ lattice of covolume U yields trade-off*

$$\alpha + \beta = 1 + \frac{k\ln 2 - \ln(k+1)}{\ln n} + o(1).$$

This is minimised over non-negative integers k at $k \in \{0,1\}$, where it equals 1 exactly. For all integers $k \geq 2$ it is strictly greater than 1.

Proof. (a) **Covolume.** The basis matrix is lower-triangular with U in position $(0,0)$ and ones on the remaining diagonal, so $\det = U$ exactly, independent of the partition and the W_i values.

(b) **Coverage.** For prime p assigned to class $\varepsilon^{(p)}$, the idempotent property gives $W_i \equiv \varepsilon_i^{(p)} \pmod{p}$ for each i, so the lattice relation $B + \sum_i b_i W_i \equiv 0 \pmod{U}$ implies $B + \sum_i \varepsilon_i^{(p)} b_i \equiv 0 \pmod{p}$, i.e. the term of S indexed by $\varepsilon^{(p)}$ is divisible by p.

(c) **Minkowski bound.** By Minkowski's First Theorem, the lattice contains a nonzero point with ℓ^∞-norm at most $C(d,N) \cdot U^{1/(k+1)}$ avoiding $N = 2^k$ forbidden hyperplanes (one per coordinate choice); the non-degeneracy of this point is guaranteed by Lemma 8.3 below.

(d) **Trade-off.** By the prime number theorem $\log U = (1+o(1))n$. Hence $\alpha = 1 - \ln(k+1)/\ln n + o(1)$ and $\beta = k\ln 2/\ln n$. Setting $f(k) = k\ln 2 - \ln(k+1)$: $f(0) = f(1) = 0$ exactly; $f'(k) = \ln 2 - 1/(k+1) = 0$ at $k+1 = 1/\ln 2 \approx 1.44$; f is convex and positive for all integers $k \geq 2$ (e.g. $f(2) = 2\ln 2 - \ln 3 \approx 0.288 > 0$). □ □

8.2 Non-degeneracy lemma

The step "there exists a nonzero lattice point avoiding the N forbidden hyperplanes" requires proof.

Lemma 8.2 (Bounded basis)**.** *Every rank-d lattice L of covolume U admits a basis $v_1, \ldots, v_d$ with $\|v_i\|_\infty \leq C_1(d) \cdot U^{1/d}$ for all i, where $C_1(d)$ depends only on d.*

This is standard: Minkowski's Second Theorem bounds $\lambda_1 \cdots \lambda_d \leq U$; classical reduction theory (see Cassels [9], Ch. VIII) gives an actual basis bounded in terms of the successive minima. We cite this without re-deriving the sharp constant; a crude bound $\|v_i\|_\infty \leq d \cdot U$ suffices for our application.

Lemma 8.3 (Non-degeneracy)**.** *Let L be a rank-d lattice of covolume U ($d \geq 2$), and let $H_1, \ldots, H_N \subset \mathbb{R}^d$ be hyperplanes through the origin. Then there exists a nonzero $v \in L$ with $\|v\|_\infty \leq N \cdot d \cdot C_1(d) \cdot U^{1/d}$ lying on none of $H_1, \ldots, H_N$.*

Proof. Fix a basis $v_1, \ldots, v_d$ from Lemma 8.2. For $c = (c_1, \ldots, c_d) \in \{0, \ldots, N\}^d$ set $v(c) = \sum_i c_i v_i$. There are $(N+1)^d - 1$ nonzero such combinations.

Fix hyperplane H_j with nonzero functional ℓ_j. Since $v_1, \ldots, v_d$ are linearly independent and $\ell_j \neq 0$, there exists m with $\ell_j(v_m) \neq 0$. Fixing the remaining $d-1$ coefficients, $v(c) \in H_j$ constrains c_m uniquely, so H_j captures at most $(N+1)^{d-1}$ tuples.

By the union bound, the number of nonzero tuples landing on at least one H_j is at most $N(N+1)^{d-1}$. Good tuples number at least

$$(N+1)^d - 1 - N(N+1)^{d-1} = (N+1)^{d-1} - 1 \geq 1 \quad (d \geq 2,\ N \geq 1).$$

Any such c^* gives $v = v(c^*) \neq 0$ with $\|v\|_\infty \leq N \cdot \max_i \|v_i\|_\infty \leq N \cdot d \cdot C_1(d) \cdot U^{1/d}$. □ □

Remark 8.4 (Well-roundedness not required). The proof requires only that ℓ_j cannot vanish on *every* basis vector simultaneously (guaranteed by linear independence), not that each basis vector individually avoids each hyperplane. If a short basis vector v_m happens to lie on H_j, its coefficient drops out of the linear equation for H_j, leaving the bound enforced through whichever other basis vector has nonzero $\ell_j(v_i)$. We verified this concretely on an adversarial example in which two of three basis vectors lay exactly on the same hyperplane: the count of bad coefficient tuples was exactly 8 out of 26 nonzero candidates, consistent with (in fact equal to) the bound $(N+1)^{d-1} - 1 = 8$ for $N = 2$, $d = 3$. The lattice need not be well-rounded.

Remark 8.5 (Hermite's constant at $k = 3$ and $k = 7$). The Minkowski bound $C_1(d) \cdot U^{1/d}$ can be sharpened using Hermite's constant γ_d: $\gamma_4 = \sqrt{2}$ (Korkine–Zolotareff [10], achieved by the D_4 lattice) and $\gamma_8 = 2$ (Viazovska [11], E_8 lattice). Substituting these exact constants at $k = 3$ or $k = 7$ sharpens the leading constant $C(k)$ without affecting the asymptotic content of Theorems 8.1–9.3.

9 A Flexible Construction: Star Chains

9.1 Construction

The *star/chain* construction uses k generators each combining only with a shared base B:

$$S = \{B,\ B + b_1,\ B + b_2,\ \ldots,\ B + b_k\}, \quad T = \{0\}.$$

This gives $k + 1$ terms (linear in k) versus 2^k for the hypercube, while the underlying lattice has the same rank $k + 1$ and covolume U.

Theorem 9.1 (Full-line achievability)**.** *Fix $\beta \in (0, 1)$. For the star/chain construction with $k = \lceil n^\beta \rceil - 1$ generators,*

$$\beta_{\text{achieved}} = \beta + o(1), \qquad \alpha_{\text{achieved}} = 1 - \beta + o(1) \quad \textit{as } n \to \infty,$$

so $\alpha + \beta = 1 + o(1)$ for every fixed $\beta \in (0, 1)$.

Proof. Covolume is U (same argument as Theorem 8.1). With $k + 1$ forbidden hyperplanes and $d = k + 1$, Lemma 8.3 gives a valid covering construction of magnitude at most $C(k) \cdot U^{1/(k+1)}$ with $C(k) = O(k)$. Then $\log(\text{magnitude}) \leq \log C(k) + n(1 + o(1))/(k + 1) = O(\log n) + \Theta(n^{1-\beta})$. For fixed $\beta < 1$, $n^{1-\beta}$ dominates $\log n$ as $n \to \infty$, giving $\alpha = 1 - \beta + o(1)$.

Caveat: convergence is pointwise in β; the rate slows as $\beta \to 1$, since the ratio $\log C(k)/n^{1-\beta}$ shrinks only as a power of $1/n$ for $\beta < 1$ but can remain $O(1)$ for β close to 1 at any finite n. □ □

9.2 Theorem 3: Global optimality of the chain structure

Definition 9.2 (Blocked hypercube–star hybrid). *For positive integers m, d with $k = md$, the (m, d)-hybrid uses m blocks of d generators each, with full d-dimensional hypercube structure* within *each block and star structure* between *blocks. This has lattice rank $1 + md = 1 + k$, covolume U, and term count $1 + m(2^d - 1)$. Setting $(m, d) = (1, k)$ recovers the pure hypercube; $(m, d) = (k, 1)$ recovers the pure star/chain.*

Theorem 9.3 (Global optimality of the chain structure). *For every $k \geq 1$ and every factorisation $k = md$ with $m, d \geq 1$,*

$$g(m, d) \ := \ \ln\bigl(1 + m(2^d - 1)\bigr) - \ln(1 + md) \ \geq \ 0,$$

with equality if and only if $d = 1$. Consequently, $\alpha + \beta - 1 = g(m, d)/\ln n + o(1)$ is minimised uniquely at $d = 1$ (the pure star/chain) and maximised uniquely at $(m, d) = (1, k)$ (the pure hypercube).

Proof. It suffices to show $2^d - 1 \geq d$ for all positive integers d, with equality only at $d = 1$.

Base: $d = 1$: $2^1 - 1 = 1 = d$. ✓

Inductive step: Suppose $2^d - 1 \geq d$. Then $2^{d+1} - 1 = 2(2^d - 1) + 1 \geq 2d + 1 > d + 1$ for $d \geq 1$.

Hence $2^d - 1 \geq d$ with strict inequality for $d \geq 2$. Since ln is monotone, this gives $g(m, d) \geq 0$ (using $m(2^d - 1) \geq md$) with equality iff $d = 1$.

Global maximum: For fixed $k = md$, $\ln(1 + md) = \ln(1 + k)$ is constant, so maximising g over factorisations of k is equivalent to maximising $f(d) := (2^d - 1)/d$. Since

$$\frac{f(d + 1)}{f(d)} = \frac{d(2^{d+1} - 1)}{(d + 1)(2^d - 1)} > 1 \quad \text{for all } d \geq 1$$

(as this simplifies to $2^d(d - 1) > -1$, trivially true), f is strictly increasing, so g is uniquely maximised at $d = k$ (forcing $m = 1$), the pure hypercube. □ □

Corollary 9.4. *The pure hypercube construction is the unique* worst *member of the blocked hypercube–star family for every fixed total budget k, and every interior point strictly interpolates between it and the chain's optimal value.*

Remark 9.5 (U-invariance and scope of Theorem 9.3). The covolume U is exactly the primorial of n for *every* configuration (m, d), every partition, and every seed. This follows algebraically from the lower-triangular basis structure and is not an assumption. We verified this computationally across 12 random partitions of varying k: U equalled the primorial in every case, even as the W_i values differed by orders of magnitude between seeds.

What Theorem 9.3 does *not* control is the *achieved* α for any specific partition, which depends on the lattice's first minimum λ_1 relative to the Minkowski bound $C \cdot U^{1/(k+1)}$. Different partitions at fixed (m, d) give different W_i values and hence different lattice geometries, with λ_1 varying accordingly — this is the Hermite-factor variance quantified in Section 9.3. The theorem compares proven *upper bounds* on α, not achieved values.

9.3 Numerical experiments and the Hermite factor

We implemented LLL lattice reduction using `mpmath` arbitrary-precision arithmetic (with precision set to the lattice bit-length plus a safety margin of 80 bits), supplemented by a BKZ-style block enumeration pass over windows of four consecutive basis vectors [15]. Plain `float64` arithmetic was found to silently produce wrong results: at 576-bit lattice entries, all useful precision is lost under a 53-bit mantissa, yielding confident but meaningless output.

Results at $n = 400$. At $k = 19$ ($\beta \approx 0.500$, aiming for the balanced point $\alpha = \beta = 1/2$), eight independent random partition seeds gave implied exponent $1 + \max(\alpha, \beta)$ with mean 1.638 and standard deviation 0.019 (range $[1.602, 1.666]$). At $k = 25$ (five seeds), the mean was 1.612 (sd 0.023, range $[1.581, 1.665]$).

The Hermite-factor framing. The asymptotic convergence to exponent 3/2 as $\beta \to 1/2$ is *proven unconditionally* by Theorem 9.1 and does not depend on these experiments. The experiments measure our specific algorithm's performance relative to the Minkowski-predicted magnitude $\log_2 U/(k+1)$. At $k = 19$ this ratio was 2.305 ± 0.255; at $k = 25$ it was 2.573 ± 0.347. The ratio did *not* improve from $k = 19$ to $k = 25$: the decrease in the raw exponent reflects the theorem's own shrinking target, not the algorithm getting relatively closer. Closing the Hermite-factor gap would require a qualitatively stronger reduction algorithm (larger BKZ blocks, true enumeration or sieving), not simply larger k.

10 Seven Converging Obstructions

Table 1 summarises the principal approaches. Seven structurally independent mathematical traditions all converge at or above $\alpha + \beta \approx 1$, each for a distinct and individually well-understood reason.

The paper's own combinatorial impossibility result (Lemma 6.3 of [1]) and our probabilistic floor (Section 5) independently confirm the same barrier. A direct test of the $c = 1$ edge case explicitly flagged as unresolved in [1] found the achieved offset

Approach	Mathematical tradition	Achieved $\alpha + \beta$	Verdict
Naive bucketing	Elementary number theory	$= 1$ exactly	Baseline
Lemma 6.2 CRT/CF	CRT + continued fractions	≈ 1, small savings	Best practical
Discrete-log sieve	Fermat / multiplicative orders	> 1 (worse)	Ruled out
First-moment prob. method	Erdős probabilistic method	$\to 1$	Confirms barrie
AP-structured randomness	Structured probability	> 1	Ruled out
Dispersion (2nd moment)	Linnik's method	Wrong sign	Mechanistically inapp
Hypercube lattice	Minkowski / geometry of numbers	$= 1$ exactly	**Proved**
Star/chain	Addition-chain theory	$= 1 + o(1)$ for all β	**Proved**

Table 1: Summary of approaches and their obstruction results.

never within an order of magnitude of the necessary theoretical floor across all tested configurations, providing empirical (not formal) evidence against that sub-case.

11 Conclusion

This investigation did not resolve the Umans–Wang (α, β)-Divisor Conjecture. It produced three new proved results: Theorems 8.1, 9.1, and 9.3, together giving an exact characterisation of, and global optimality result for, the blocked hypercube–star family of lattice constructions. Most substantively, it produced a multi-method triangulation of a single barrier $\alpha + \beta \approx 1$, approached independently through combinatorics, probability theory, Diophantine approximation, discrete logarithms, and the geometry of numbers, with each approach failing for a distinct and well-understood reason.

11.1 A calibrated self-assessment

Theorems 8.1 and 9.3 rest on elementary calculus and a one-line inductive inequality; both were independently re-derived and numerically verified. Theorem 9.1 is correct as a pointwise-in-β asymptotic statement, with the slowing convergence rate near $\beta \to 1$ identified explicitly. The non-degeneracy lemma (Lemma 8.3) is now fully proved by an elementary pigeonhole argument, verified computationally and on a worked adversarial example; it rests on one cited classical fact (Lemma 8.2) rather than a fully self-contained derivation.

The numerical results of Section 9.3 characterise our specific reduction algorithm's Hermite-factor gap to a proven asymptotic target; the achieved-to-predicted ratio did not improve with k over the tested range, so these experiments should not be read as independent evidence of convergence to exponent $3/2$. The literature search supporting novelty claims was targeted, not exhaustive. Most importantly, this document has not been reviewed by anyone outside the conversation in which it was produced. Its appropriate use is as a careful starting point for expert review, not as an established result.

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